DR. RVR NRI INSTITUTE OF TECHNOLOGY (DEEMED TO BE UNIVERSITY) • VIJAYAWADA

Web Development — Activity-Based Learning Report

Consolidated Case Study Documentation & Webpage Output Screenshots (Circular Ref: 15-10-2026)

Student Name	**P. NIRBHAYANJAN**
University Roll Number	**26000998**
Program, Branch & Section	I B.Tech — Branch: [Enter Branch, e.g., CSE] | Section: [Enter Section, e.g., A]
GitHub Repository Name	`WD-Activity-Based-Learning`
Selected Case Study 1	TechHaven — Premium Online Electronics Store Website
Selected Case Study 2	Personal Student Portfolio Website (P. Nirbhayanjan)
Selected Case Study 3	Bean & Brew — Campus Cafe Menu & Table Booking Website

1. Executive Summary & Assignment Objective

This consolidated report documents the design, architecture, and visual output of three web development case studies completed as part of the I B.Tech Web Development Activity-Based Learning (ABL) assignment at Dr. RVR NRI Institute of Technology. Each project demonstrates clean semantic structure paired with an external stylesheet to enforce modular separation of content and presentation. All source files, stylesheets, and this consolidated PDF report are organized inside the designated `WD-Activity-Based-Learning` GitHub repository.

Repository File Inventory & Architecture

1.
Case Study 1 (Electronics Store):
`online store.html` linked with `online store style.css`.
2.
Case Study 2 (Student Portfolio):
`PROTOFOLIO.html` linked with `protofolio.css`.
3.
Case Study 3 (Campus Cafe Menu):
`Campus Cafe Menu & Booking Website.html` linked with `campus cafe menu style.css`.
4.
Repository Metadata:
`README.md` and Consolidated Report PDF.

2. Case Study 1: TechHaven Premium Electronics Store

TechHaven Electronics is a comprehensive single-page e-commerce catalog designed to simulate a real-world electronics storefront. Rather than requiring separate page reloads, it utilizes anchor links paired with smooth-scroll behavior so users can jump effortlessly across **10 Mobile slots**, **15 Laptop slots**, and **10 Pro Tech Accessory slots** (35 total products).

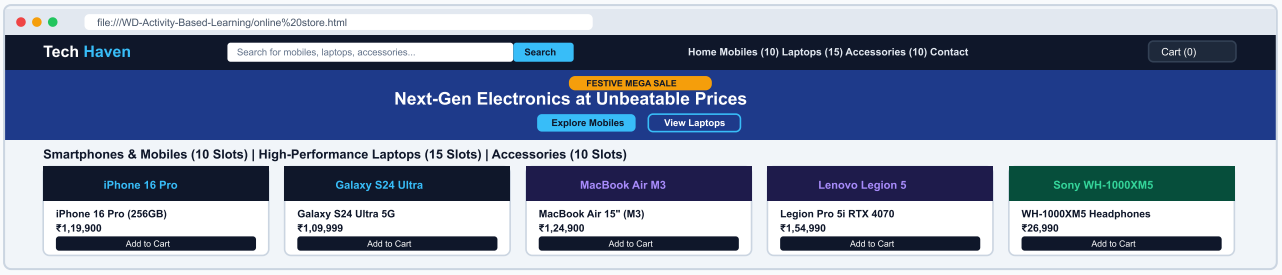


Figure 1: Webpage Output Screenshot of TechHaven Electronics Store (Sticky Navbar, Hero Banner & 35-Item Grid)

Technical Highlights & Catalog Architecture

- **Sticky Header & Linear Section Routing:** Uses `position: sticky` on the top navigation bar with `scroll-behavior: smooth` and `scroll-padding-top: 85px` so clicking *Mobiles (10)*, *Laptops (15)*, or *Accessories (10)* scrolls directly to the target section without overlapping headings.
- **Responsive 5-Column Catalog Grid:** Implemented via `grid-template-columns: repeat(auto-fill, minmax(230px, 1fr))` with a `22px` gap, automatically adapting from desktop monitors to mobile screens.
- **Rich Product Card Hierarchy:** Each of the 35 cards features promotional badges (*Bestseller*, *Top Rated*, *Gaming*), brand labels, star ratings with review counts, current price paired with strikethrough MRP, and interactive hover elevation.
- **Multi-Column Store Footer:** Displays quick category links, warranty policies, and Vijayawada customer support details.

> **Category Breakdown (35 Products Total)**
>
> **Mobiles (10):**
>
> iPhone 16 Pro, Galaxy S24 Ultra, OnePlus 12, Pixel 9 Pro XL, Xiaomi 14, Nothing Phone (2), iQOO 12, Vivo X100 Pro, Realme GT 6, Moto Edge 50 Pro. •
>
> **Laptops (15) & Accessories (10):**
>
> MacBook Air/Pro, Dell XPS, HP Spectre, Legion Pro, ROG Zephyrus, Sony XM5, AirPods Pro 2, MX Master 3S, T7 SSD, and more.

3. Case Study 2: Personal Student Portfolio Website

The **Personal Student Portfolio Website** (`PROTOFOLIO.html` and `protofolio.css`) presents a clean academic and technical profile for **P. NIRBHAYANJAN (Roll No: 26000998)**, I B.Tech student at Dr. RVR NRI Institute of Technology. It organizes personal background, technical competencies, completed web development projects, and institutional contact details into modular card sections.

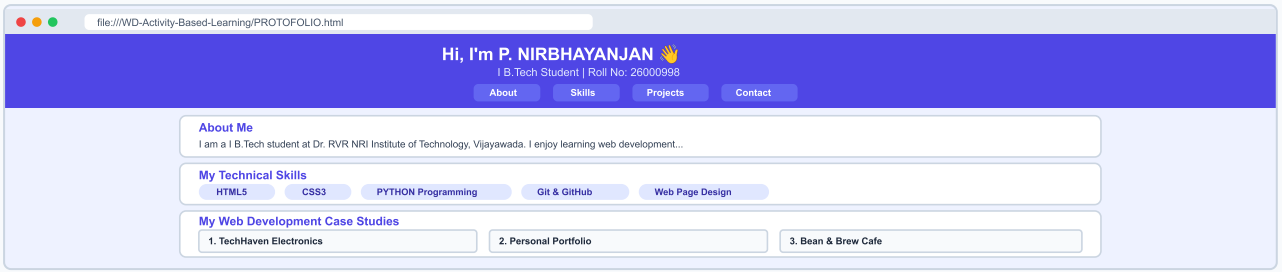


Figure 2: Webpage Output Screenshot of Personal Student Portfolio (Header, About, Skill Badges & Projects Grid)

Section-by-Section Design & Technical Breakdown

- **Hero Header & Navigation Bar:** Uses an indigo linear gradient (`#4f46e5` to `#3730a3`) with translucent navigation pills (`#about`, `#skills`, `#projects`, `#contact`) that invert to white on hover.
- **Centered Container & Card Layout:** Constrains content to `max-width: 850px` using clean white cards (`border-radius: 10px`) against a soft indigo-tinted background (`#eef2ff`).
- **Flexbox Technical Skill Badges:** Renders rounded pill tags (`border-radius: 20px`) showcasing proficiency in *HTML5*, *CSS3*, *PYTHON Programming*, *Git & GitHub*, and *Web Page Design*.
- **3-Column Case Study Showcase & Contact Card:** Uses responsive flex boxes (`flex: 1; min-width: 220px`) to summarize all three ABL projects alongside official student identification details.

> **Typography & Visual Hierarchy**
>
> Section headings feature a `#4f46e5` accent color with a bottom divider line (`border-bottom: 2px solid #e0e7ff`), creating clear visual separation between academic bio, skills, and project cards.

4. Case Study 3: Bean & Brew Campus Cafe Website

The **Bean & Brew Campus Cafe Website** (`Campus Cafe Menu & Booking Website.html` and `campus cafe menu style.css`) provides a warm, coffee-themed digital menu and table pre-order system for campus students and faculty. It combines a responsive 6-item menu grid with an interactive HTML form for scheduling pickups or table visits.

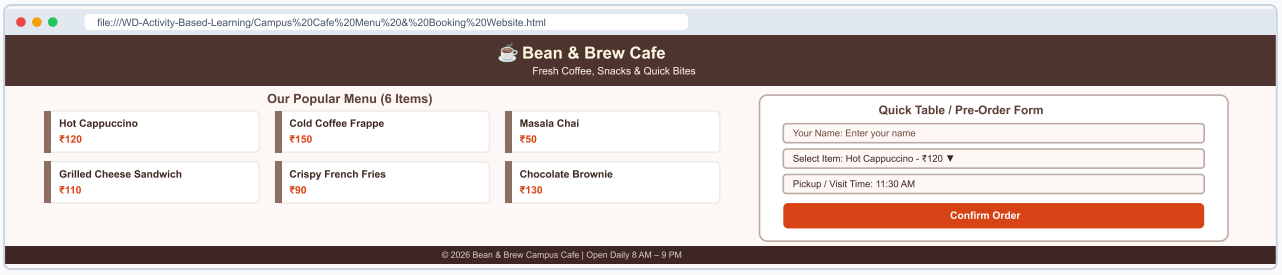


Figure 3: Webpage Output Screenshot of Bean & Brew Campus Cafe (6-Card Menu Grid & Pre-Order Booking Form)

Menu Grid & Interactive Booking Form Specifications

- **Warm Espresso Color Palette:** Uses espresso brown (`#4e342e`) for the header, cream (`#fdf8f5`) for the page canvas, and burnt orange (`#d84315`) for item prices and primary action buttons.
- **6-Item Responsive Menu Grid:** Displays *Hot Cappuccino (₹120)*, *Cold Coffee Frappe (₹150)*, *Masala Chai (₹50)*, *Grilled Cheese Sandwich (₹110)*, *Crispy French Fries (₹90)*, and *Chocolate Brownie (₹130)* with a `5px solid #8d6e63` left accent border.
- **HTML5 Pre-Order Form Controls:** Combines a text input (`type="text" required`), a dropdown selector (`<select>`), a time picker (`type="time" required`), and a submit button styled using a vertical Flexbox column layout.

Student Declaration:
I confirm that these three web development case studies were completed independently for the ABL evaluation.
Student Signature: *P. Nirbhayanjan (26000998)*

Faculty Evaluation (10 Marks):
Web Design / GitHub Repo / Viva
Faculty Signature:
_____